

Модуль ядра с init

Файл test1_init.ko

```
#include <linux/module.h>
#include <linux/kernel.h>

int init_module(void)
{
    pr_info("My module loaded!");

    return 0;
}

void cleanup_module(void)
{
    pr_info("My module unloaded!");
}

MODULE_LICENSE("GPL");
```

Makefile

```
obj-m += test1_init.o

all:
    make -C /lib/modules/6.17.0-35-generic/build M=$(PWD) modules

clean:
    make -C /lib/modules/6.17.0-35-generic/build M=$(PWD) clean
```

1. Компилирование модуля

make

```
● tair@tair-Bravo-15-C7VE:~/Eltex/lesson20$ make
make -C /lib/modules/6.17.0-35-generic/build M=/home/tair/Eltex/lesson20 modules
make[1]: вход в каталог «/usr/src/linux-headers-6.17.0-35-generic»
make[2]: вход в каталог «/home/tair/Eltex/lesson20»
warning: the compiler differs from the one used to build the kernel
The kernel was built by: x86_64-linux-gnu-gcc-13 (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
You are using:          gcc-13 (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
CC [M] test1_init.o
MODPOST Module.symvers
WARNING: modpost: missing MODULE_DESCRIPTION() in test1_init.o
CC [M] test1_init.mod.o
CC [M] .module-common.o
LD [M] test1_init.ko
BTF [M] test1_init.ko
Skipping BTF generation for test1_init.ko due to unavailability of vmlinux
make[2]: выход из каталога «/home/tair/Eltex/lesson20»
make[1]: выход из каталога «/usr/src/linux-headers-6.17.0-35-generic»
```

2. Загрузка и удаление модуля

`sudo insmod test1_init.ko`

`sudo rmmod test1_init`

```
[ 3121.55277] My module unloaded!  
tair@tair-Bravo-15-C7VE:~/Eltex/lesson20$ sudo dmesg | tail -n2  
[ 3122.284232] My module loaded!  
[ 3141.595277] My module unloaded!  
tair@tair-Bravo-15-C7VE:~/Eltex/lesson20$
```

Модуль ядра с переопределёнными read/write

Файл test2_rw.c

```
#include <linux/module.h>  
#include <linux/kernel.h>  
#include <linux/fs.h>  
#include <linux/mutex.h>    // вместо rwlock.h  
#include <linux/uaccess.h>  // опционально, simple_read_from_buffer уже в  
fs.h  
  
static int major = 0;  
static DEFINE_MUTEX(lock); // замена rwlock_t на mutex  
static char test_string[15] = "Hello!\n\0";  
  
ssize_t test_read(struct file *fd, char __user *buff, size_t size, loff_t *off)  
{  
    size_t rc;  
  
    if (mutex_lock_interruptible(&lock))  
        return -ERESTARTSYS;  
  
    rc = simple_read_from_buffer(buff, size, off, test_string, 15);  
  
    mutex_unlock(&lock);  
    return rc;  
}  
  
ssize_t test_write(struct file *fd, const char __user *buff, size_t size, loff_t *off)  
{  
    size_t rc = 0;  
  
    if (size > 15)  
        return -EINVAL;  
  
    if (mutex_lock_interruptible(&lock))
```

```

        return -ERESTARTSYS;

rc = simple_write_to_buffer(test_string, 15, off, buff, size);

mutex_unlock(&lock);
return rc;
}

static struct file_operations fops = {
    .owner = THIS_MODULE,
    .read = test_read,
    .write = test_write,
};

int init_module(void)
{
    pr_info("My module 2 loaded!\n");

    major = register_chrdev(0, "test2_rw", &fops);
    if (major < 0)
        return major;

    pr_info("Major number is %d\n", major);
    return 0;
}

void cleanup_module(void)
{
    unregister_chrdev(major, "test2_rw");
    pr_info("My module 2 unloaded!\n");
}

MODULE_LICENSE("GPL");

```

Makefile

```

obj-m += test2_rw.o

all:
    make -C /lib/modules/6.17.0-35-generic/build M=$(PWD) modules

clean:
    make -C /lib/modules/6.17.0-35-generic/build M=$(PWD) clean

```

1. Компилирование модуля

make

```
• tair@tair-Bravo-15-C7VE:~/Eltex/lesson20/part2$ make
make -C /lib/modules/6.17.0-35-generic/build M=/home/tair/Eltex/lesson20/part2 modules
make[1]: вход в каталог «/usr/src/linux-headers-6.17.0-35-generic»
make[2]: вход в каталог «/home/tair/Eltex/lesson20/part2»
warning: the compiler differs from the one used to build the kernel
The kernel was built by: x86_64-linux-gnu-gcc-13 (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
You are using:          gcc-13 (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
CC [M] test2_rw.o
test2_rw.c:11:9: warning: no previous prototype for 'test_read' [-Wmissing-prototypes]
  11 | ssize_t test_read(struct file *fd, char __user *buff, size_t size, loff_t *off)
      |          ^~~~~~
test2_rw.c:22:9: warning: no previous prototype for 'test_write' [-Wmissing-prototypes]
  22 | ssize_t test_write(struct file *fd, const char __user *buff, size_t size, loff_t *off)
      |          ^~~~~~
MODPOST Module.symvers
WARNING: modpost: missing MODULE_DESCRIPTION() in test2_rw.o
CC [M] test2_rw.mod.o
CC [M] .module-common.o
LD [M] test2_rw.ko
BTF [M] test2_rw.ko
Skipping BTF generation for test2_rw.ko due to unavailability of vmlinux
make[2]: выход из каталога «/home/tair/Eltex/lesson20/part2»
make[1]: выход из каталога «/usr/src/linux-headers-6.17.0-35-generic»
```

2. Загрузка модуля в ядро

make insmod test2_rw.c

```
• tair@tair-Bravo-15-C7VE:/dev$ sudo dmesg | tail -n5
[ 5316.521345] Major number is 504
[ 5350.851214] [UFW BLOCK] IN=en01 OUT= MAC=01:00:5e:00:00:01:c0:06
[ 5353.850553] [UFW BLOCK] IN=en01 OUT= MAC=01:00:5e:00:00:fb:a4:2b
[ 5382.850719] [UFW BLOCK] IN=en01 OUT= MAC=01:00:5e:00:00:01:c0:06
[ 5385.849936] [UFW BLOCK] IN=en01 OUT= MAC=01:00:5e:00:00:fb:a4:2b
```

3. Создание специального файла устройства

```
• tair@tair-Bravo-15-C7VE:/dev$ sudo mknod test2_rw c 504 0
• tair@tair-Bravo-15-C7VE:/dev$ ls -la test2_rw
crw-r--r-- 1 root root 504, 0 июн 14 05:56 test2_rw
• tair@tair-Bravo-15-C7VE:/dev$
```

4. Чтение и запись

```
• tair@tair-Bravo-15-C7VE:/dev$ cat test2_rw
Hello!
• tair@tair-Bravo-15-C7VE:/dev$ echo "123" > test2_rw
bash: test2_rw: Отказано в доступе
• tair@tair-Bravo-15-C7VE:/dev$ sudo chmod +666 test2_rw
• tair@tair-Bravo-15-C7VE:/dev$ echo "123" > test2_rw
• tair@tair-Bravo-15-C7VE:/dev$ cat test2_rw
123
o!
```